CLASSWORK 10

Be Bayesian

CPE 486/586

Instructor: Rahul Bhadani

Due: October 19, 2026, In-Class

20 points

Last month, a small engineering office logged the 100 emails that reached its mail server. A staff member read them and labeled the messages spam ($Y = 1$) or awesome ($Y = 0$), where awesome means legitimate mail. The filter tracks two binary features.

- $X_1 = 1$ when the email contains the word "offer", and $X_1 = 0$ otherwise.
- $X_2 = 1$ when the email contains the word "attached", and $X_2 = 0$ otherwise.

Table 1 lists the counts. Read it one row at a time. The row $(1, 0)$, for instance, says that 12 spam emails and 18 awesome emails contained "offer" without "attached".

Table 1. Counts of the 100 labeled emails.

X_1 ("offer")	X_2 ("attached")	Spam ($Y = 1$)	awesome ($Y = 0$)	Row total
1	1	24	6	30
1	0	12	18	30
0	1	0	30	30
0	0	4	6	10
Column total		40	60	100

Part 1: Priors and Likelihoods from Counts (4 points)

(a) **[1 point]** Estimate the priors from the column totals of Table 1.

$$P(Y=1) = \underline{\quad 40/100 = 0.4 \quad} \qquad P(Y=0) = \underline{\quad 60/100 = 0.6 \quad}$$

(b) **[2 points]** Fill in Table 2 with the remaining class-conditional joint likelihoods $P(X_1 = x_1, X_2 = x_2 \mid Y = y)$. Divide a count by its class total. The spam column must add to 1, and so must the awesome column.

Table 2. Joint likelihoods.

x_1	x_2	$P(x_1, x_2 \mid Y=1)$	$P(x_1, x_2 \mid Y=0)$
1	1	$24/40 = 0.6$	$6/60 = 0.1$
1	0	$12/40 = 0.3$	$18/60 = 0.3$
0	1	$0/40 = 0$	$30/60 = 0.5$
0	0	$4/40 = 0.1$	$6/60 = 0.1$
Column sum		1.0	1.0

(c) **[1 point]** Apply the sum rule to Table 2 and fill in Table 3. As a pattern, $P(X_1 = 1 \mid Y = y) = P(1,1 \mid Y = y) + P(1,0 \mid Y = y)$.

Table 3. Marginal likelihoods of the two words.

	$y = 1$ (spam)	$y = 0$ (awesome)
$P(X_1 = 1 \mid Y = y)$	$0.6 + 0.3 = 0.9$	$0.1 + 0.3 = 0.4$
$P(X_2 = 1 \mid Y = y)$	$0.6 + 0 = 0.6$	$0.1 + 0.5 = 0.6$

Solution

Look at the second row of Table 3. The word "attached" appears in 60% of spam and in 60% of awesome. Keep that in mind for Part 5(b).

Part 2: Full Bayes with the Joint Likelihood (5 points)

Two new emails arrive. Email A contains "offer" and lacks "attached", so $\mathbf{x}_A = (1,0)$. The second one, Email B, lacks "offer" and contains "attached", so $\mathbf{x}_B = (0,1)$. In this part you'll take the likelihoods straight from Table 2.

(a) **[4 points]** Complete Table 4.

Table 4. Posteriors under the joint model.

Quantity	Email A, $\mathbf{x}_A = (1,0)$	Email B, $\mathbf{x}_B = (0,1)$
$P(\mathbf{x} \mid Y=1)\,P(Y=1)$	$0.3 \times 0.4 = 0.12$	$0 \times 0.4 = 0$
$P(\mathbf{x} \mid Y=0)\,P(Y=0)$	$0.3 \times 0.6 = 0.18$	$0.5 \times 0.6 = 0.30$
$P(\mathbf{x})$ by the sum rule	$0.12 + 0.18 = 0.30$	$0 + 0.30 = 0.30$
$P(Y=1 \mid \mathbf{x})$	$0.12/0.30 = 0.4$	$0/0.30 = 0$
$P(Y=0 \mid \mathbf{x})$	$0.18/0.30 = 0.6$	$0.30/0.30 = 1$
Decision ω^* (spam or awesome)	awesome	awesome
$P(\text{error} \mid \mathbf{x})$	$1 - 0.6 = 0.4$	$1 - 1 = 0$

(b) **[1 point]** Count the emails in Table 1 whose features equal $\mathbf{x}_A$ and divide by 100. Compare the result with $P(\mathbf{x}_A)$ in Table 4, then explain the agreement in one sentence.

Observed frequency of $\mathbf{x}_A$: $(12+18)/100 = 0.30$

Solution

The two values agree exactly. A joint likelihood is a column frequency, so $P(\mathbf{x} \mid Y=y)\,P(Y=y) = \dfrac{n(\mathbf{x},y)}{n_y} \cdot \dfrac{n_y}{100} = \dfrac{n(\mathbf{x},y)}{100}$, and the sum rule then adds the row of Table 1. The joint model reproduces the training data with no approximation.

Part 3: Naive Bayes (5 points)

Naive Bayes replaces the joint likelihood with a product of marginals,

$$P(\mathbf{x} \mid Y=y) \approx P(X_1 = x_1 \mid Y=y)\,P(X_2 = x_2 \mid Y=y).$$

Take the factors from Table 3. When a word is absent, use $P(X_j = 0 \mid Y=y) = 1 - P(X_j = 1 \mid Y=y)$.

(a) **[4 points]** Complete Table 5 for Email A and Email B.

Table 5. Posteriors under naive Bayes.

Quantity	Email A, $\mathbf{x}_A = (1,0)$	Email B, $\mathbf{x}_B = (0,1)$
$P(x_1 \mid Y=1)\,P(x_2 \mid Y=1)$	$0.9 \times 0.4 = 0.36$	$0.1 \times 0.6 = 0.06$
$P(x_1 \mid Y=0)\,P(x_2 \mid Y=0)$	$0.4 \times 0.4 = 0.16$	$0.6 \times 0.6 = 0.36$
Spam term, times $P(Y=1)$	$0.36 \times 0.4 = 0.144$	$0.06 \times 0.4 = 0.024$
awesome term, times $P(Y=0)$	$0.16 \times 0.6 = 0.096$	$0.36 \times 0.6 = 0.216$
$P_{\text{NB}}(\mathbf{x})$ by the sum rule	$0.144 + 0.096 = 0.24$	$0.024 + 0.216 = 0.24$
$P(Y=1 \mid \mathbf{x})$	$0.144/0.24 = 0.6$	$0.024/0.24 = 0.1$
$P(Y=0 \mid \mathbf{x})$	$0.096/0.24 = 0.4$	$0.216/0.24 = 0.9$
Decision ω^* (spam or awesome)	spam	awesome
$P(\text{error} \mid \mathbf{x})$	$1 - 0.6 = 0.4$	$1 - 0.9 = 0.1$

(b) **[1 point]** Write down the class that the joint model picks for Email A and the class that naive Bayes picks for it.

Joint model: awesome (0.4) Naive Bayes: spam (0.6)

Solution

The data and the prior stayed fixed between Table 4 and Table 5. Only the independence assumption changed, and it moved Email A across the 0.5 threshold.

Part 4: Testing the Naive Assumption (2 points)

(a) **[1 point]** Conditional independence requires

$$P(X_1 = 1, X_2 = 1 \mid Y = y) = P(X_1 = 1 \mid Y = y)\,P(X_2 = 1 \mid Y = y).$$

Check this equation for $y = 1$, then for $y = 0$, with numbers from Tables 2 and 3.

Class	Left side (Table 2)	Right side (Table 3)	Equal?
$y = 1$ (spam)	0.6	$0.9 \times 0.6 = 0.54$	No
$y = 0$ (awesome)	0.1	$0.4 \times 0.6 = 0.24$	No

(b) **[1 point]** Compare $P_{\text{NB}}(\mathbf{x}_A)$ in Table 5 with the observed frequency from Part 2(b). Explain the gap in one or two sentences.

Solution

(a) In spam the two words appear together more often than independence predicts ($0.6 > 0.54$). awesome shows the opposite pattern ($0.1 < 0.24$). The assumption fails in the two classes.
(b) Naive Bayes builds its own distribution over $\mathbf{x}$ out of the marginals, $P_{\text{NB}}(\mathbf{x}) = \sum_y P(Y = y) \prod_j P(x_j \mid Y = y)$. Part 4(a) showed dependence within a class, so this product table departs from Table 2, and its evidence of 0.24 departs from the observed 0.30. A student who checks $P_{\text{NB}}(\mathbf{x})$ against the counts can detect the misspecification before reading a single posterior.

Part 5: Interpretation (4 points)

(a) **[1 point]** The joint model reports $P(Y = 1 \mid \mathbf{x}_B) = 0$. Which count in Table 1 produced this zero? Would you deploy a filter that is certain Email B is awesome? Say which posterior for Email B you'd report, and give a reason.

(b) **[1 point]** Email C contains "offer" and "attached", so $\mathbf{x}_C = (1, 1)$. Compute $P(Y = 1 \mid \mathbf{x}_C)$ under the joint model and under naive Bayes.

Joint: $\dfrac{0.24}{0.24 + 0.06} = 0.8$ Naive: $\dfrac{0.216}{0.216 + 0.144} = 0.6$

Naive Bayes gives Email C the posterior it gave Email A. Use Table 3 to explain why.

(c) **[1 point]** Suppose the office installs the same filter in a department where spam makes up 10% of incoming mail, so $P(Y = 1) = 0.1$. Keep the likelihoods of Table 2. Recompute $P(Y = 1 \mid \mathbf{x}_C)$ under the joint model and state the new decision.

$$P(Y = 1 \mid \mathbf{x}_C) = \frac{0.6 \times 0.1}{0.6 \times 0.1 + 0.1 \times 0.9} = \frac{0.06}{0.15} = 0.4$$ Decision: awesome

(d) **[1 point]** Bayes' rule in odds form reads

$$\frac{P(Y = 1 \mid \mathbf{x})}{P(Y = 0 \mid \mathbf{x})} = \frac{P(Y = 1)}{P(Y = 0)} \times \frac{P(\mathbf{x} \mid Y = 1)}{P(\mathbf{x} \mid Y = 0)}.$$

With the original prior $P(Y = 1) = 0.4$, write the posterior odds for Email C under the joint model as prior odds times likelihood ratio. Then evaluate the sigmoid at $z = \ln(\text{posterior odds})$ and compare the result with Part 5(b).

Prior odds: $0.4/0.6 = 2/3$

Likelihood ratio: $0.6/0.1 = 6$

Posterior odds: $(2/3) \times 6 = 4$

$g(\ln(\text{posterior odds}))$: $1/(1 + 1/4) = 0.8$

Solution

(a) The zero comes from the spam count in row $(0,1)$. In the spam column, 0 emails contain "attached" without "offer". Forty messages cannot rule that pattern out, and a phishing email with a fake invoice "attached" fits it well. If such an email arrives, the joint model gives it zero probability of being spam, and the cross-entropy loss $-\log P(Y=1 \mid \mathbf{x})$ becomes infinite. I'd report the naive value of 0.1. Laplace smoothing, $(n(\mathbf{x},y)+1)/(n_y+4)$, is the usual repair, and it gives $1/44 \approx 0.023$ for the empty cell.

(b) Table 3 gives $P(X_2=1 \mid Y=1) = P(X_2=1 \mid Y=0) = 0.6$. The factor 0.6 multiplies the spam term and the awesome term alike, so it cancels in Bayes' rule, and the naive posterior depends on X_1 alone. Taken by itself, "attached" carries zero evidence. The joint model sees the interaction that the marginals hide. "Offer" with "attached" points to spam (0.8), and "offer" alone points to awesome (0.4).

(c) The likelihood ratio stays at 6 in favor of spam. Prior odds fall to $0.1/0.9 = 1/9$, so the posterior odds become $6/9 = 2/3 < 1$, and the decision changes to awesome. A filter carries the prior of the office that trained it. The department's own mail should set $P(Y)$, in the spirit of "today's posterior is tomorrow's prior."

(d) $g(\ln 4) = 1/(1+e^{-\ln 4}) = 1/(1+\frac{1}{4}) = 0.8$, the joint posterior of Part 5(b). The sigmoid turns log posterior odds into a probability. Logistic regression sets $z = \boldsymbol{\theta}^\top \mathbf{x}$ and fits $\boldsymbol{\theta}$ by maximum likelihood, so it estimates the log posterior odds directly and skips separate estimates of $P(Y)$ and $P(\mathbf{x} \mid Y)$.

Bonus: From Naive Bayes to Logistic Regression (+4 points, optional)

For binary features, naive Bayes produces log posterior odds that are linear in the features,

$$\ln \frac{P(Y=1 \mid \mathbf{x})}{P(Y=0 \mid \mathbf{x})} = w_0 + w_1 x_1 + w_2 x_2, \qquad w_j = \ln \frac{p_{j1}}{p_{j0}} - \ln \frac{1-p_{j1}}{1-p_{j0}},$$

where $p_{jy} = P(X_j = 1 \mid Y = y)$.

(i) Show that $w_2 = 0$ for this data set.

(ii) Compute the joint-model posterior for $\mathbf{x} = (0,0)$. Then argue that a logistic regression with $z = \theta_0 + \theta_1 x_1 + \theta_2 x_2$ cannot reproduce the joint-model posteriors of the four feature patterns. Hint: compare the change in log odds when x_1 switches on with $x_2 = 0$, and again with $x_2 = 1$.

Solution

(i) $w_2 = \ln(0.6/0.6) - \ln(0.4/0.4) = 0$. For reference, $w_1 = \ln(0.9/0.4) - \ln(0.1/0.6) = \ln 13.5$ and $w_0 = \ln(2/3) + \ln(0.1/0.6) + \ln(0.4/0.4) = \ln(1/9)$. The sigmoid then returns 0.6 when $x_1 = 1$ and 0.1 when $x_1 = 0$, in agreement with Table 5.

(ii) For $(0,0)$ the joint model gives $0.04/(0.04+0.06) = 0.4$. The joint log odds are $\ln(2/3)$ at $(0,0)$, $\ln(2/3)$ at $(1,0)$, $-\infty$ at $(0,1)$, and $\ln 4$ at $(1,1)$. With $x_2 = 0$, switching x_1 on changes the log odds by 0. Switching x_1 on with $x_2 = 1$ produces an infinite change, and the change stays near 4.7 after Laplace smoothing. A linear z forces the same change θ_1 in the two cases, so it cannot fit. The model needs the product feature $x_1 x_2$. Naive Bayes and plain logistic regression share the linear form here, and that form is what misses the interaction.